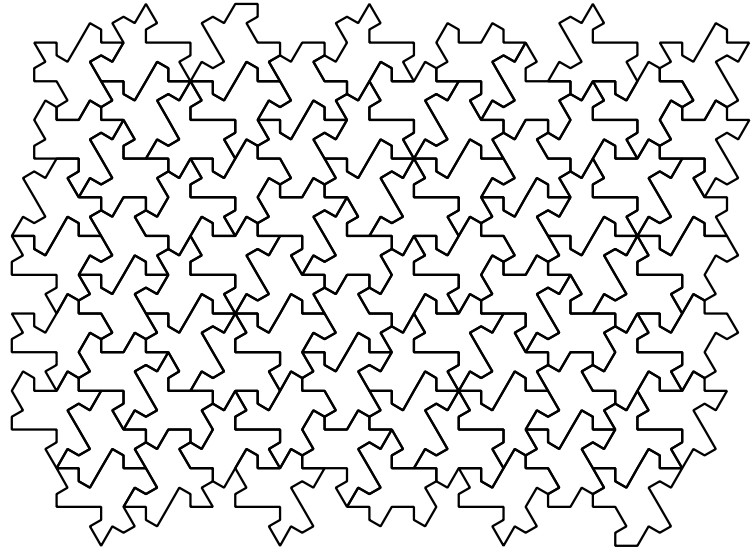


Match a tiling.

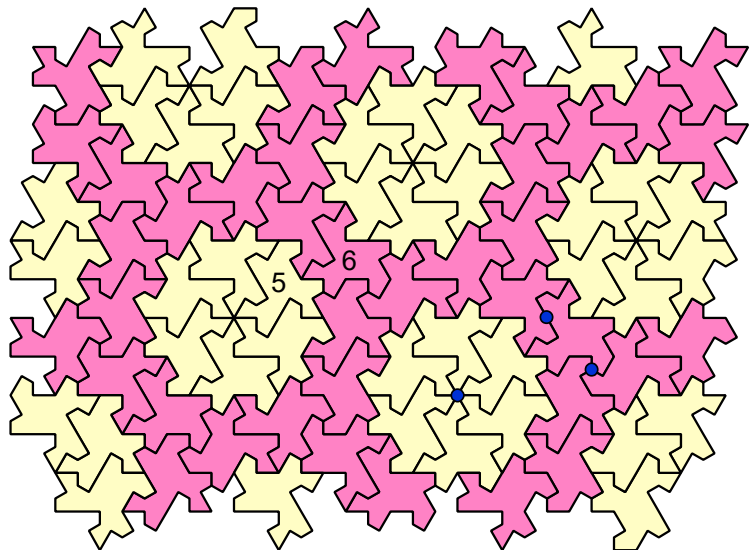
Say that you start with a 2-isohedral monohedral tiling. (Such as Joseph Myers has on his website.

<https://www.polyomino.org.uk/mathematics/polyform-tiling/>

The one at the right is a polykite with size $n = 14$ (and $k=2$, for 2-isohedral)

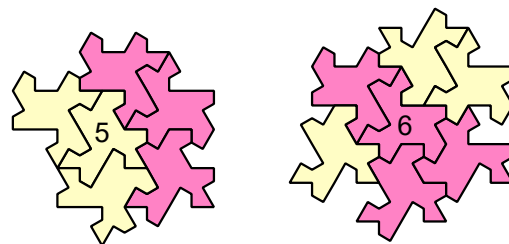


Then shade each orbit a different color. This helps to see the structure. I marked the rotation centers with blue dots. We have rotation centers of order 6, 3 and 2. There are no reflection lines of symmetry. The orbifold is “632.” We also count how many neighbor tiles each orbit has. I labeled a yellow tile “5” and a pink tile “6”.



orbifold = 632 (centers marked with blue dots)

Looking at the neighbors surrounding each tile, we look at how many of the neighbors are the opposite orbit. Each of these has three neighbors of the opposite orbit. We can use this information to find which tiling we are dealing with.



$\begin{smallmatrix} 5 \\ 3 \end{smallmatrix}$

$\begin{smallmatrix} 6 \\ 3 \end{smallmatrix}$

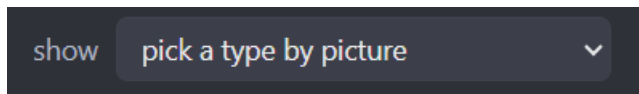
When the app opens you can see some boxes that show $5_2 6_2$.



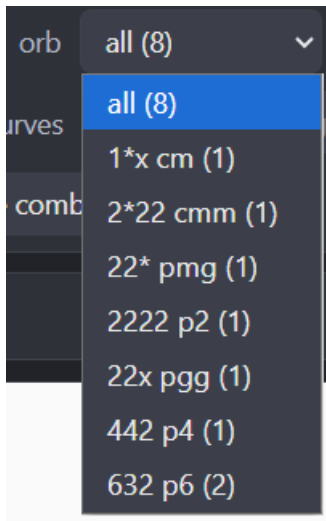
We need to change this to read $5_3 6_3$.



Set the mode to “Pick a type by picture.”

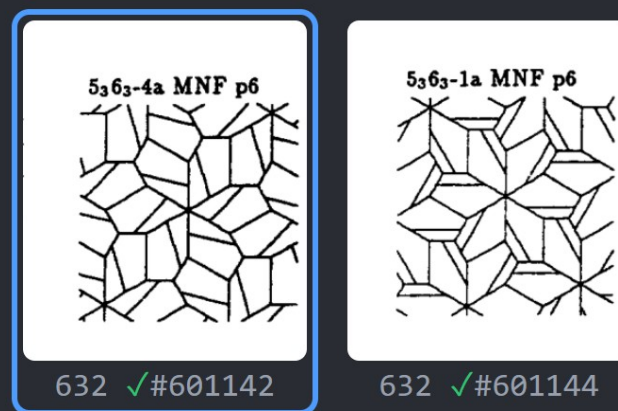


Here you would see 8 examples in the panel on the right. But we can narrow it further with the orbifold. Click the dropdown menu near “orb” and select 632.



This leaves us with just two pictures to choose from in the panel. Now we have to study the pictures given and figure out which one matches our tiling. In this case, we either have a 5 sided figure at the 6 fold rotation center or a 6 sided figure. Our tiling has the 5 sided figure, so it is type #601142. (Note that the app has a short cut where you can enter this type number immediately, if you know it.)

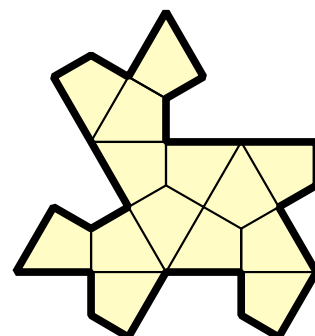
2 types match **632**. Click a figure to load it.



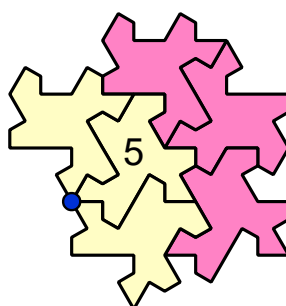
These shapes are polykites, so we can view them as composed of kite figures. Count the number of edges around the perimeter to get 24.

Set the size to 24.

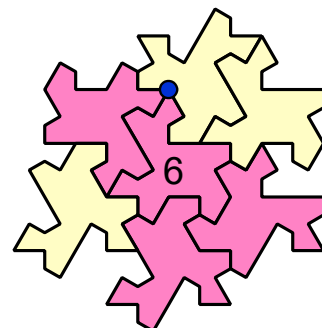
sides 24 ▾



Next, see how many edges are in each side. (I'm calling a edge of a kite an “edge” and the boundary between two tiles a “side”) So for the yellow orbit, starting at the blue dot and going around the tile clockwise, we have 8 edges in the first side, then 4 edges, followed by 2 and 2 and 8. Now find where that blue dot must be on the pink tile. For the pink orbit we have 4,2,4,4,2,8. Please note that both must add up to 24. If the orientation was opposite, we would go around the other direction.



Clockwise
8,4,2,2,8



Clockwise
4,2,4,4,2,8

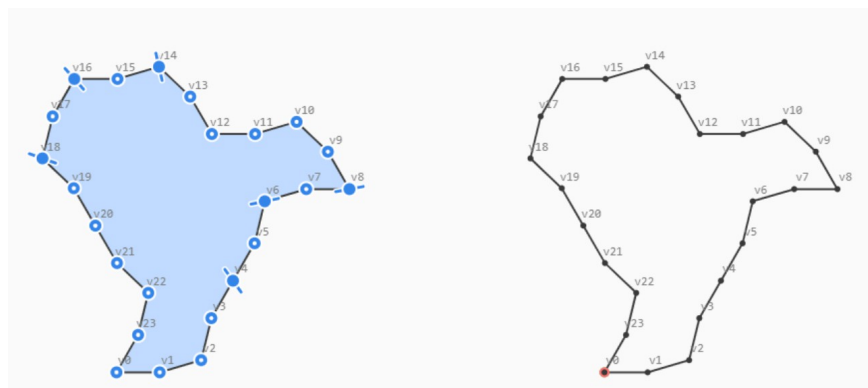
Set the mode to “find combo by side sizes.”

show find combo by side sizes ▾

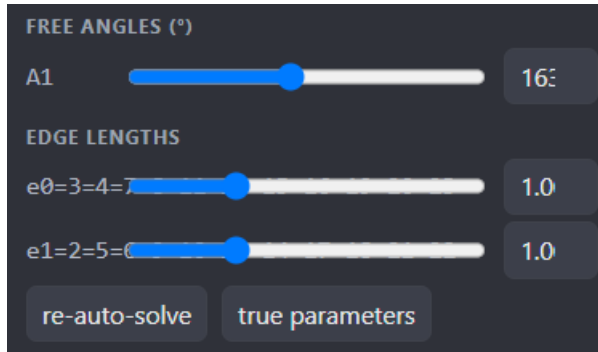
In the box for sides per orbit enter the lists, separated by a slash. Then click the “Show” button.

8,4,2,2,8 / 4,2,4,4,2,8 anchor ☒ show

The right panel shows options. Here we have only one option. Click it and you'll see a shape.

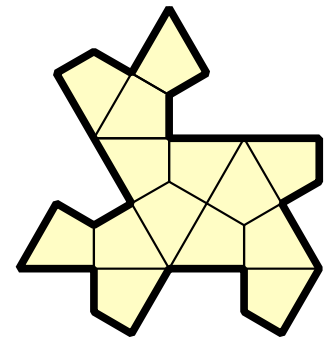


Note that the panel has an angle slider (A1) and a couple of length ratios.

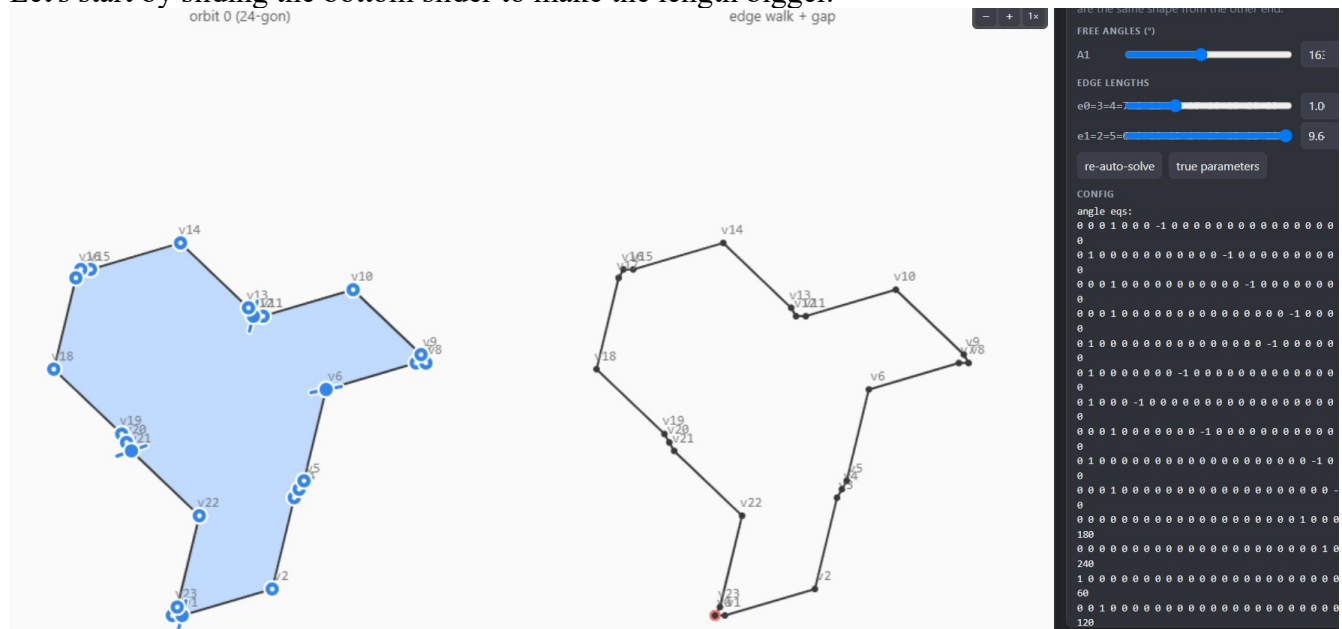


When we move these raw sliders, often the shape will break apart and not close. Then we have to adjust other sliders. There is a re-auto-solve button to make this easier.

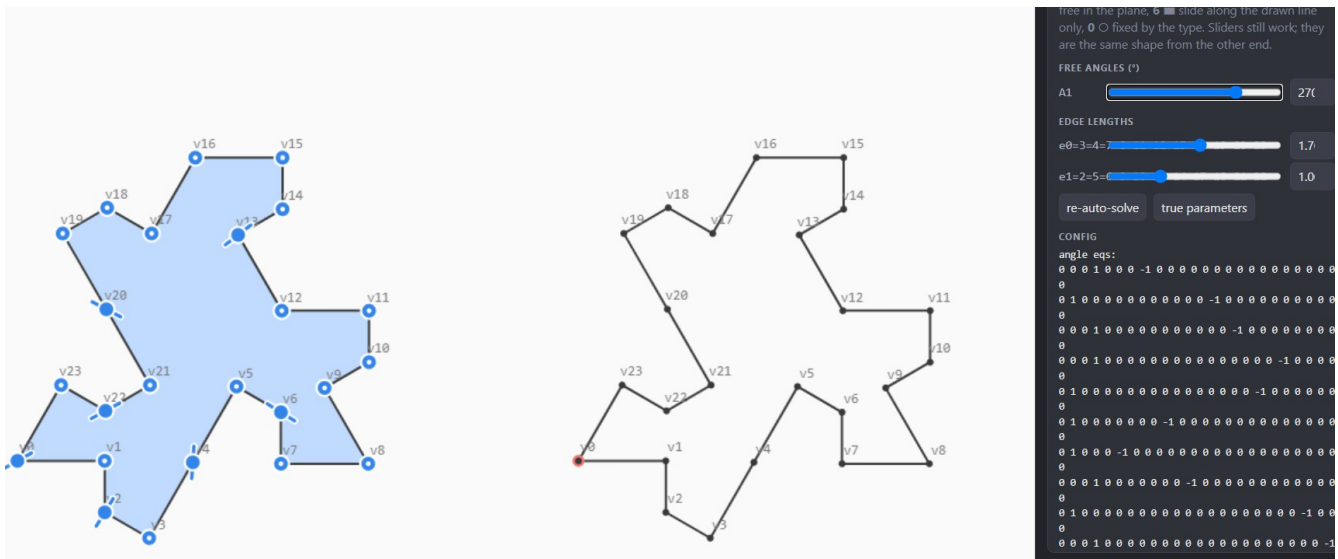
We are aiming at the figure at the left. Note how it has a couple of edges that are in a straight line. Also how some edges are a bit longer than others. In fact, since we are dealing with a 30-60-90 triangle, we know that the edge ratios are 1 to the square root of 3. So the ratio needs to be about 1:1.732 with two long edges lined up in two places.



Let's start by sliding the bottom slider to make the length bigger.

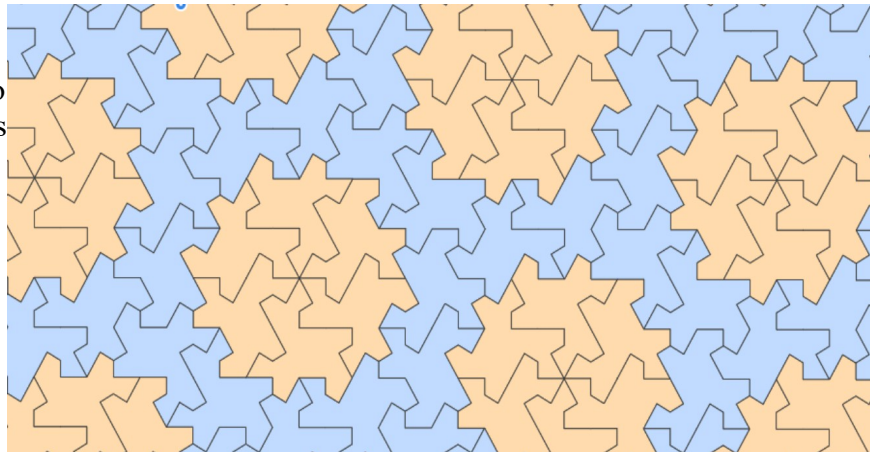


This is the wrong one to make longer. We can see now there are a couple of places where two short edges are in a straight line. We need two long edges in a straight line. So move the bottom slider back to 1 and move the middle slider to about 1.732

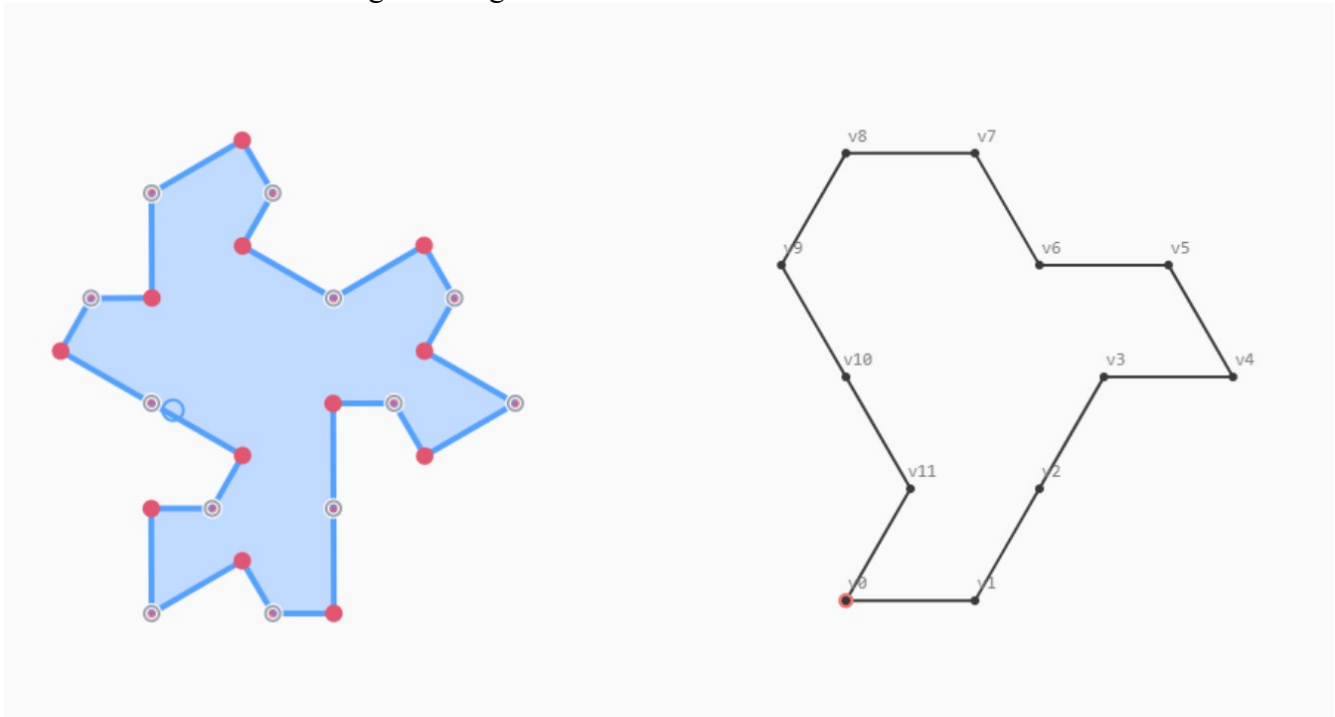


When we get to 270 degrees, this looks like what we are aiming at. We can switch to full tiling mode to see what it all looks like. This looks like what we wanted.

We can go a step further. Note that up above in green that it says “closes – reducible.” That means that we can find this tiling with fewer edges.



Hit the “reduce” button. We get this figure below.



Notice this is the same shape, but now some of the vertices are colored red. These are curve points. Try dragging one a bit and you can change the curve. If you click and drag on an edge away from a control point, you will add another control point. Here's a random shape to illustrate it.

